Extending a Simulation Framework for Performance Assessment of Parallel Applications

Abstract

DisCostiC is a cross-architecture simulation framework that simulates massively parallel applications by taking the performance properties of the hardware-software interaction into account. The main novelty of this analytic-model-based simulator is its full-scale scope; it is essentially an automated version of the analytical ab-initio white- or gray-box models on the full hierarchy of parallel systems, including cores, chips, nodes, networks, clusters, their individual inherent bottlenecks and the interactions among them. To this end, the simulations employ the standard Roofline and Hockney models as well as their enhanced ECM and LogGP family models.

In order to simulate an application, a *skeleton* or *blueprint* of the code must be constructed. The skeleton is expressed as a tailored domain-specific embedded language (DSEL) program, which describes the performance-relevant features of the program in an abstract way. To ease skeleton construction, a compact and intuitive annotation language helps to facilitate the semi-automatic production of application blueprints via static analysis.

The full-scale scope of the simulator enables reproducing the interplay of complex parallel programs and different components of current supercomputers. It further allows for model-based design-space exploration of hypothetical parallel programs on future high-performance systems. Simulations are performed in a well-controlled environment without requiring massive resources for computations and data transfers and are faster than traditional approaches to MPI simulation since *no code is executed* on real target systems. The simulated traces can be visualized using the Chromium web browser or the Vampir tool.

The master's thesis involves the following tasks

- Getting familiar with the DisCostiC simulation framework and reproducing the current straightforward test cases.
- Construction of DSEL code for a near-production CFD application, performance assessment of the simulator predictions and their comparisons against actual application runs
- Validation of simulator predictions using metrics beyond runtime - data volume, communication time, other fundamental and derived metrics
- Framework extension that incorporates a threading and energy model to simulate the energy and performance characteristics of highly parallel threaded applications.
- Nature of work: Theory (20%), Conception (40%), Implementation (40%)

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